A34312 No Calculator

UNIVERSITY OF BIRMINGHAM

School of Mathematics

Programmes in the School of Mathematics	Second Examination
Programmes involving Mathematics	Second Examination
Programmes in the School of Mathematics	Final Examination
Programmes involving Mathematics	Final Examination

Multivariable and Vector Analysis

Level I 2MVA 06 25667

Level H 2MVA3 06 35172

May/June Examinations 2024–25

Three Hours

Full marks will be obtained with complete answers to all FOUR questions. Each question carries equal weight. You are advised to initially spend no more than 45 minutes on each question and then to return to any incomplete questions if you have time at the end. An indication of the number of marks allocated to parts of questions is shown in square brackets.

No calculator is permitted in this examination.

1. (a) Find the direction of the tangent to the intersection of the two surfaces

$$xy + yz + zx = 5 \text{ and } xyz = 2$$

at the point (1, 1, 2). [6]

(b) Consider the function

$$f(x,y,z) = x + y + z + e^{xyz}.$$

Calculate

(i) The gradient of the function at the point (0, 1, 1).

(ii) The direction and rate of steepest descent for the function at the point (0, 1, 1).

(iii) The directional derivative of the function at the point (0, 1, 1) in the direction from the point (0, 1, 1) to the point (-2, -1, 2).

[7]

(c) Using Lagrange multipliers, find the maximum and minimum values of the function

$$f(x,y,z) = x + 2y + 3z$$

subject to the constraint

$$x^2 + y^2 + z^2 = 14.$$

[12]

2. (a) Consider the following double integral

$$I = \iint_D \frac{x}{x^2+y^2}\mathrm{d}x\mathrm{d}y,$$

where D is the triangular region with vertices at (0, 0), (1, 0) and (1, 1) in Cartesian coordinates.

(i) Express the double integral as repeated integrals in polar coordinates, integrating with respect to r first. There is no need to evaluate the integral in this step.

(ii) Express the double integral as repeated integrals in polar coordinates, integrating with respect to θ first. Again, there is no need to evaluate the integral in this step.

(iii) Calculate the integral using the results obtained in (i) or (ii) or otherwise.

[10]

(b) Consider the region S, which is inside the spherical surface $x^2+y^2+z^2=4$ but outside the cylindrical surface $x^2+y^2=1$.

(i) Express the volume V of the region as repeated integrals using cylindrical coordinates.

(ii) Express the volume V of the region as repeated integrals using spherical coordinates.

(iii) Calculate the volume V using either of the two coordinate systems, using the results obtained in (i) or (ii) or otherwise.

[15]

3. (a) Consider the force $\vec{F} = (\sin(y)e^x - py)\hat{\imath} + (\cos(y)e^x - p)\hat{\jmath}$, $p \in \mathbb{R}$, and find the work W when a particle is moved by the force $\vec{F}$ once around a closed contour C formed by the x-axis and the upper semi-circle $(x–a)^2 + y^2 = a^2, y \geq 0,\ a > 0$, where C is oriented counterclockwise. [8]

(b) Consider $\vec{r} = (x, y, z)$ and let $\vec{r}_0$ be the position vector of a point $P_0 = (x_0, y_0, z_0)$. Calculate, showing all steps in your solution, $\nabla \cdot \left(\dfrac{\vec{q}}{q}\right)$ in the domain $\mathbb{D}$, where $\vec{q} = \vec{r} - \vec{r}_0$, $q = |\vec{q}|$, and $\mathbb{D} = \mathbb{R}^3 \backslash \{P_0\}$. [9]

(c) Consider the vector field $\vec{F} = 2\sin(x)yz\hat{\imath} + x^3e^z\hat{\jmath} - \cos(x)yz^2\hat{k}$ and find, showing all steps, the flux of $\vec{F}$ through the surface S of the cube $0 \leq x \leq 1$, $0 \leq y \leq 1$, $0 \leq z \leq 1$, where a normal vector to the surface S is pointing outwards. [8]

4. (a) Consider a circle of radius $R = 1$ centred at the origin in the (x, y)-plane. A particle moves along the circle at a constant speed. Prove that the acceleration vector and the position vector of the particle are collinear at any point of the circle. [5]

(b) Evaluate, showing all steps in your solution, the integral $\int_C y\,\mathrm{d}s$, where C is an arc of the curve $y = \sqrt{2x}$ between the points $(0,0)$ and $(1, \sqrt{2})$. [7]

(c) Demonstrate, showing all steps, that the vector field $\vec{F} = (2x + y^2 z)\hat{\imath} + 2xyz\hat{\jmath} + (xy^2 - 1)\hat{k}$ is a gradient field. Hence, find the potential function f by computing line integrals. [13]

A34312

Multivariable and Vector Analysis

Do not complete the attendance slip, fill in the front of the answer book or turn over the question paper until you are told to do so.

Important Reminders

- Coats and outer-wear should be placed in the designated area.
- Unauthorised materials (e.g., notes or Tippex) **MUST** be placed in the designated area.
- Check that you **DO NOT** have any unauthorised materials with you (e.g., in your pockets or in your pencil case).
- Mobile phones and smart watches **MUST** be switched off and placed in the designated area or under your desk. They must not be left on your person or in your pockets.
- You are **NOT** permitted to use a mobile phone as a clock. If you have difficulty in seeing a clock, please alert an Invigilator.
- You are **NOT** permitted to have writing on your hand, arm or other body part.
- Check that you do not have writing on your hand, arm or other body part; if you do, you must inform an Invigilator immediately.
- Alert an Invigilator immediately if you find any unauthorised item upon you during the examination.

Any students found with non-permitted items upon their person during the examination, or who fail to comply with Examination rules, may be subject to the Student Conduct procedures.